MineNetCD: A Benchmark for Global Mining Change Detection on Remote Sensing Imagery

Weikang Yu, *Graduate Student Member, IEEE*, Xiaokang Zhang, *Senior Member, IEEE*, Richard Gloaguen, Xiao Xiang Zhu, *Fellow, IEEE*, and Pedram Ghamisi, *Senior Member, IEEE*

***Abstract*— Monitoring land changes triggered by mining activities is crucial for industrial control, environmental management, and regulatory compliance, yet it poses significant challenges due to the vast and often remote locations of mining sites. Remote sensing technologies have increasingly become indispensable to detect and analyze these changes over time. We thus introduce MineNetCD, a comprehensive benchmark designed for global mining change detection using remote sensing imagery. The benchmark comprises three key contributions. First, we establish a global mining change detection dataset featuring more than 70k paired patches of bitemporal high-resolution remote sensing images and pixel-level annotations from 100 mining sites worldwide. Second, we develop a novel baseline model based on a change-aware fast Fourier transform (ChangeFFT) module, which enhances various backbones by leveraging essential spectrum components within features in the frequency domain and capturing the channelwise correlation of bitemporal feature differences to learn change-aware representations. Third, we construct a unified change detection (UCD) framework that currently integrates 20 change detection methods. This framework is designed for streamlined and efficient processing, using the cloud platform hosted by HuggingFace. Extensive experiments have been conducted to demonstrate the superiority of the proposed baseline model compared with 19 state-of-the-art change detection approaches. Empirical studies on modularized backbones comprehensively confirm the efficacy of different representation learners on change detection. This benchmark represents significant advancements in the field of remote sensing and change detection, providing a robust resource for future research and applications in global mining monitoring. Dataset and Codes are available via the link.**

***Index Terms*— Benchmark, frequency-domain learning, mining change detection, remote sensing, unified framework.**

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Weikang Yu is with the Chair of Data Science in Earth Observation, Technical University of Munich (TUM), 80333 Munich, Germany, and also with Helmholtz Institute Freiberg for Resource Technology, Helmholtz-Zentrum Dresden-Rossendorf, 09599 Freiberg, Germany (e-mail: w.yu@hzdr.de).

Xiaokang Zhang is with the School of Information Science and Engineering, Wuhan University of Science and Technology, Wuhan 430081, China (e-mail: natezhangxk@gmail.com).

Richard Gloaguen is with Helmholtz Institute Freiberg for Resource Technology, Helmholtz-Zentrum Dresden-Rossendorf, 09599 Freiberg, Germany.

Xiao Xiang Zhu is with the Chair of Data Science in Earth Observation and Munich Center for Machine Learning, Technical University of Munich (TUM), 80333 Munich, Germany.

Pedram Ghamisi is with Helmholtz Institute Freiberg for Resource Technology, Helmholtz-Zentrum Dresden-Rossendorf, 09599 Freiberg, Germany, and also with the Lancaster Environment Centre, Lancaster University, LA1 4YR Lancaster, U.K. (e-mail: p.ghamisi@hzdr.de).

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I. Introduction

MINING operations are crucial to provide the materials required for infrastructure, technology, and well-being and ensure the energy transition but have significant environmental impacts [1]. Open-pit mine areas, where minerals are extracted from a pit in the ground, induce large excavations and produce massive wastes in the form of tailings and waste piles during their life cycle [2], [3]. These activities are closely linked to adverse impacts such as ecosystem fragmentation, degradation, geotechnical disasters, and biodiversity loss [4]. Land degradation is a primary concern due to mining-induced alterations such as excavations and the removal of topsoil and vegetation. This results in soil erosion, landscape instability, and ecosystem disruption [5], [6]. Sustainability aims to minimize environmental impact and promote a better relationship between humans and our planet. While mining is inherently not sustainable, metal sourcing can be part of a sustainable society. Sourcing metals sustainably will require considering well-being as a purpose and preserving natural capital [7]. To achieve sustainability, sourcing the raw resources that our societies increasingly demand will necessitate transparent and inclusive stakeholder participation, as well as a comprehensive understanding of the consequences of extractive activities [8].

Geospatial data collected through satellite, drone, and aerial imagery play a vital role in achieving metal sourcing sustainability by providing real-time, detailed perspectives on mining sites [9], [10], [11], [12]. Tools such as machine learning and artificial intelligence (AI) help analyze these data to enhance the accuracy and speed of data analysis by transforming raw geospatial data into actionable insights [13]. This improved analysis enables better anticipation and mitigation of environmental risks, thereby reducing mining's environmental footprint and aligning with the goals of AI for social good in building responsible AI [14]. Nowadays, remote sensing change detection technology using multitemporal images has become a key tool in tracking changes in the landscape caused by mining operations [15], [16], [17].

Mining change detection aims to quantify the changes caused by mining operations on the Earth's surface by comparing the multitemporal images acquired in the same geographical areas [18], [19]. It is essentially a dense labeling task for pixelwise classification. In recent years, deep-learning-based methods have emerged as the dominant approach for change detection, leveraging encoder–decoder structures based on convolutional neural networks (CNNs) and

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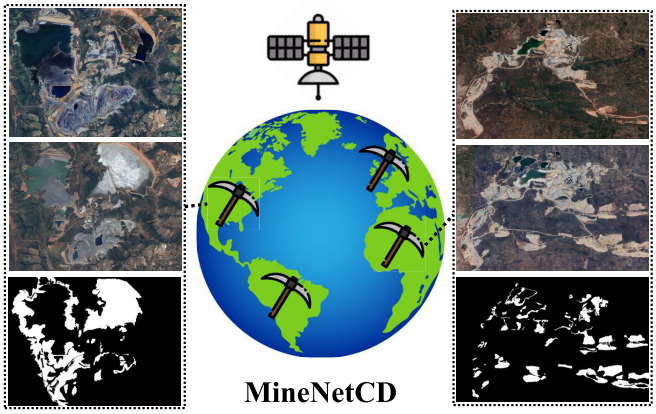


Fig. 1. MineNetCD benchmark: a comprehensive framework for monitoring global mining developments using remote sensing imagery.

transformers [20], [21], [22], [23]. Nevertheless, the success of these models heavily relies on large-scale, high-quality training datasets, posing challenges in terms of data annotation and diversity. Li et al. [16] constructed the first open-pit mine change detection dataset, OMCD, with a resolution of 2 m containing bitemporal images from 39 open-pit mining sites in China. On this basis, a CNN-based change detection method for open-pit mines has been investigated, in which the multiscale difference features of bitemporal images were extracted and exploited by a Siamese network in an encoder–decoder structure [15]. To address the issue of data scarcity, Xie et al. [17] proposed a novel subinstance data augmentation method based on a generative adversarial network to generate realistic and diverse samples.

For the change detection model, various strategies have been proposed to improve the feature representation of CNNs by leveraging multiscale feature interaction and aggregation [24], [25], [26], [27]. On this basis, attention mechanisms have been applied to refine feature maps along the spatial and channel dimensions [28], [29]. To handle the limitations of global context modeling in CNNs, recent methods have introduced transformers into change detection [30], [31]. They either take transformer as their backbone or integrate it with CNNs to capture global contextual information [32], [33], [34]. Seeking to combine the strengths of CNNs and transformers, attention-based multiscale transformer networks have been investigated to extract global–local contextual information [35], [36], [37]. Despite these innovations, efficiently modeling long-range contextual dependencies while preserving local details remains a challenge, especially in mining change detection. Tasks in this domain are shaped by land-cover changes from mining operations, which often manifest as irregularly shaped alterations and display spectral variability on remote sensing images. These complexities heighten the challenges for change-aware representation learning, in stark contrast to the more predictable shapes observed in urban areas. In addition, differentiating true changes from pseudochanges calls for an enhanced exploration of bitemporal feature correlations to effectively use essential components within the feature maps. Thus, mining change detection necessitates exceptionally precise models that can accurately delineate the irregular edges of mining-induced patterns.

Open science has emerged as a vital paradigm in the development of AI applications for Earth Observation (AI4EO), emphasizing transparency and the sharing of scientific research, data, and methodologies with the broader community. This involves not only disseminating final research outcomes but also sharing underlying data, models, and codes. To streamline these processes, several open-source change detection frameworks[1,2] have been introduced, aiming to facilitate the application of various change detection methodologies on widely used datasets [38], [39]. However, these frameworks often require additional dataset preprocessing and sharing pretrained models remains cumbersome, leading to complex and repetitive operations. In addition, the availability of datasets and trained model weights is limited, making it challenging to reproduce the results of different approaches. Consequently, current frameworks still demand a high level of specialized skills, setting a relatively high bar for researchers interested in this field.

In this article, we attempt to build a benchmark for global mining change detection and to prompt the achievement of mining sustainability goals. The main contributions of this article are summarized as follows.

1) The global mining change detection dataset contains more than 70k paired patches of bitemporal high-resolution remote sensing images and pixel-level annotations acquired from 100 mining sites worldwide.
2) We developed a novel baseline model based on change-aware fast Fourier transform (ChangeFFT). It aids various backbones in leveraging key spectrum components within the features and efficiently capturing the channelwise correlation of bitemporal features to learn discriminative change-aware representations.
3) A unified change detection (UCD) framework is constructed by integrating 20 advanced change detection methods. It is designed for a more streamlined and efficient process, leveraging the cloud platform HuggingFace.

This article is organized as follows. Section II describes the proposed mining change detection dataset. After that, a new baseline model and a new change detection framework are introduced in Sections III and IV, respectively. The experimental performance is evaluated in Section V. Finally, Section VI concludes this article.

II. MineNetCD Dataset

Using AI and deep learning for open-pit mining detection from remote sensing images has demonstrated its potential to enhance the precision of open-pit mining detection under various environmental conditions. Nevertheless, the success of these models heavily relies on large-scale, high-quality training datasets [22], posing challenges in terms of data annotation and diversity. The main challenges in constructing

[1]https://github.com/walking-shadow/Simple-Remote-Sensing-Change-Detection-Framework

[2]https://github.com/likyoo/open-cd

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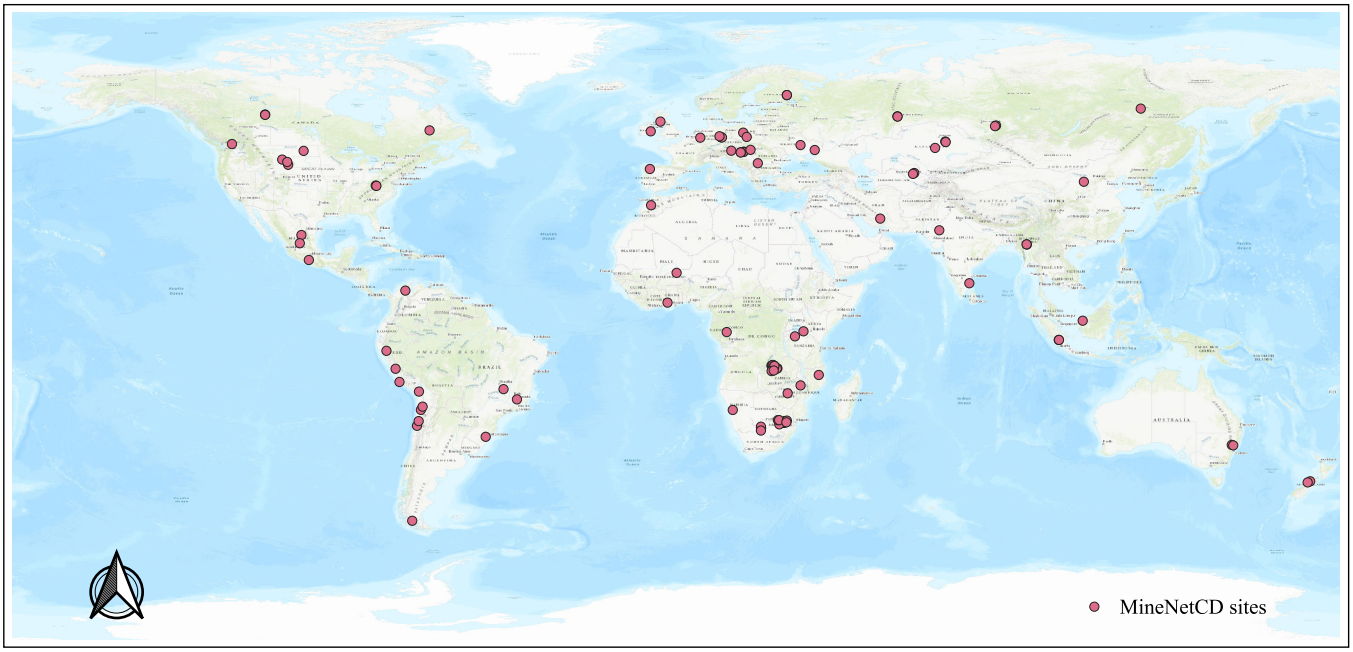


Fig. 2. Global spatial distribution of 100 mining sites from the MineNetCD dataset, spanning six continents. These sites exhibit various land-cover conditions, mining sizes, occurrence times, spatial distributions, and phenology states, resulting in considerable spectral heterogeneity in remote sensing imagery. The basemap is from ArcGIS Online.

a global mining change detection dataset are twofold. First, it is difficult to accurately locate mining sites in countries worldwide, hindering image acquisition. Second, compared with other scenes, dense labeling of mining changes is more challenging due to the spectral similarity between mining areas and other ground objects, as well as the irregular shapes and fragmented boundaries of changed areas.

We sought to present a global high-resolution open-pit mining change detection dataset, MineNetCD. It has been recorded that more than 6000 active mining sites exist globally [3]. We intentionally selected 100 representative mining sites from different geographical regions to build the benchmark. The spatial distribution of mining sites in the MineNetCD dataset is illustrated in Fig. 2. The sites were chosen based on changes in the mining area and their spatial distribution. We aimed to capture representative mining locations from around the world. By overlaying the mining polygons onto remote sensing images, we can precisely locate the mining sites and retrieve the desired images, as shown in Fig. 3. This dataset has the following properties.

1) *The Largest:* MineNetCD covers a wide range of open-pit mining sites located in different countries and regions across six continents. Each site involves bitemporal high-resolution imagery pairs with a spatial resolution of 1.2 m captured through Google Earth Engine service. MineNetCD consists of more than 70k paired patches that have a total coverage of approximately 6756.88 km^2.
2) *High Heterogeneity:* The dataset comprises bitemporal image pairs capturing mining sites of various sizes, shapes, and industries. As shown in Fig. 4(d), the spectral intensity histograms of changed and unchanged pixels between preevent and postevent imageries reveal significant spectral heterogeneity and intensity variations across different temporal instances and locations. This high-diversity dataset enables a robust evaluation of change detection models' generalization performance. In addition, Fig. 5(a) illustrates the number of mining sites per country, highlighting South Africa, Congo, and the USA as having the highest counts, with South Africa leading significantly. Fig. 5(b) quantifies the impacted land areas due to mining activities. Canada emerges as

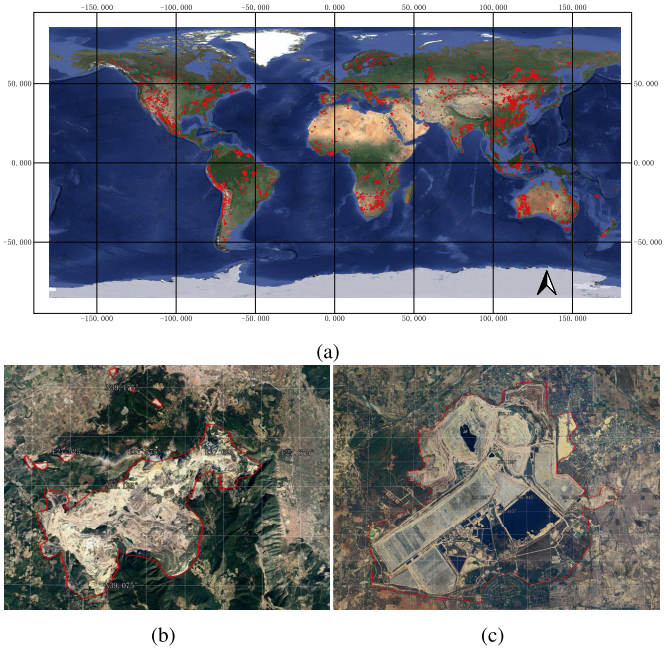


Fig. 3. To locate the mining sites. (a) Overlay the global mining polygon data [3] onto Google Earth. This allows the sites to be identified. (b) and (c) Two such examples.

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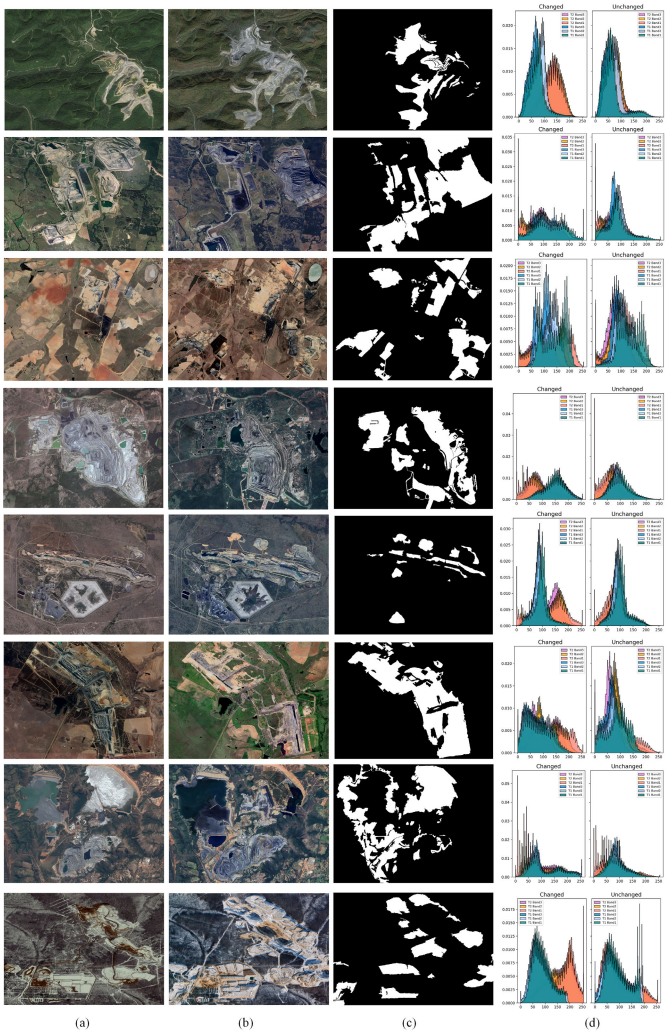


Fig. 4. Typical samples in MineNetCD. (a) and (b) Preevent and postevent imageries. (c) Change maps. (d) Spectral intensity histograms of changed and unchanged pixels between preevent and postevent imageries.

the most affected, with an extensive 80 km^2 of altered landscape, followed by South Africa and Congo.

3) *Toward Mining-Driven Changes:* Unlike much of the existing research, which primarily focuses on urban [28], [40], building-related [41], [42], agricultural [43], and natural disaster-induced changes [44], as illustrated in Table II, this study addresses land use/cover changes driven by mining operations, with a particular emphasis on industrial expansion and its associated environmental impacts, such as deforestation and erosion. Since land changes caused by mining are not frequent, current mining monitoring schemes typically adopt temporal intervals ranging from a few years to several decades [15], [45], [46]. Based on our observations of actual changes at each mining site, we selected varying time intervals for different mines to ensure that significant change patterns could be detected between bitemporal images. As shown in Fig. 5(c), the average time interval between bitemporal images in the MineNetCD dataset is approximately 89.4 months (around seven years), with a minimum interval of four years to guarantee a discernible change pattern in the mining areas. The acquisition times for the prechange images range from 2009 to 2018, while the postchange images were captured between 2018 and 2023.

4) *Fine-Grained Data:* Semiautomatic labeling is used for image annotations using the eCognition developer software. First, the segmentation map is created by co-segmenting the stacked bitemporal images using the multiresolution segmentation algorithm to generate spatially corresponding segments. Subsequently, image interpretation experts manually label the changed areas by comparing the pre- and postchange images. This process ultimately yields pixelwise annotations with fine-grained boundaries. More details can be observed in Fig. 4.

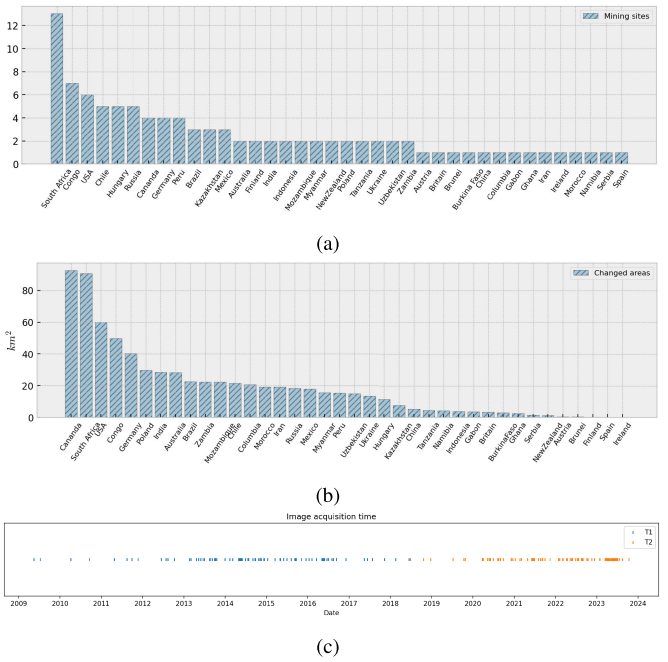


Fig. 5. Statistics of MineNetCD. (a) No. of mining sites in each country. Changed areas of (b) each country and (c) each site. (d) Image acquisition time.

III. MineNetCD Model

This section provides the details of the proposed MineNetCD model, as shown in Fig. 6. First, the bitemporal images are forwarded into a modularized Siamese encoder with two weight-shared backbones to extract the multilevel bitemporal deep features, which are then subtracted into feature differences. After that, a ChangeFFT module is incorporated to enhance the multilevel feature differences into change-aware representations with a frequency-domain learning strategy. Consequently, the enhanced feature maps are processed by a UperNet-based change decoder to generate a change map, in which the multilevel features are progressively processed and upsampled with a pyramid pooling module (PPM) and a feature pyramid network (FPN).

A. Modularized Siamese Encoder

In the literature, encoder–decoder-based change detection models typically use a fixed Siamese encoder to extract deep

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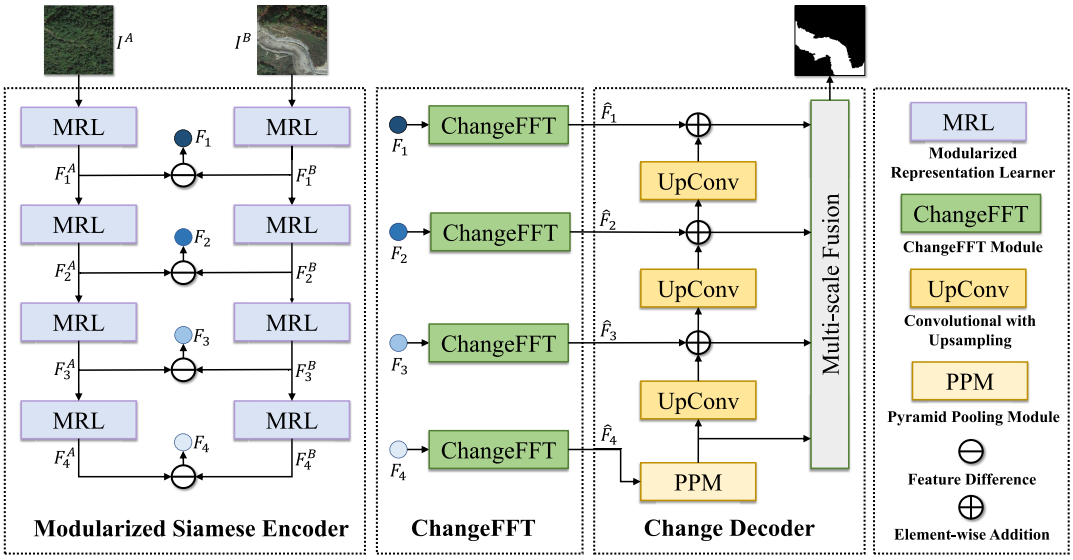


Fig. 6. Framework of the proposed MineNetCD model. First, bitemporal deep features are extracted at multiple levels by a modularized Siamese encoder and subtracted to obtain feature differences. Next, the ChangeFFT module amplifies these multilevel feature differences into change-aware representations through frequency-domain learning. Finally, a change decoder reconstructs a pixelwise change map by fusing multilevel change-aware representations.

TABLE I
OVERVIEW OF THE BACKBONE MODELS ADOPTED IN THIS STUDY

Backbone Models	Learning Strategy	#Parameters	FLOPs
ResNet Series			
ResNet-18	CNN	11M	1.8G
ResNet-50		24M	3.8G
ResNet-101		43M	7.6G
Swin Transformer Series			
Swin-T	Transformers	28M	4.5G
Swin-S		50M	8.7G
Swin-B		88M	15.4G
VMamba Series			
VMamba-T	Mamba	30M	4.8G
VMamba-S		50M	8.7G
VMamba-B		89M	15.4G

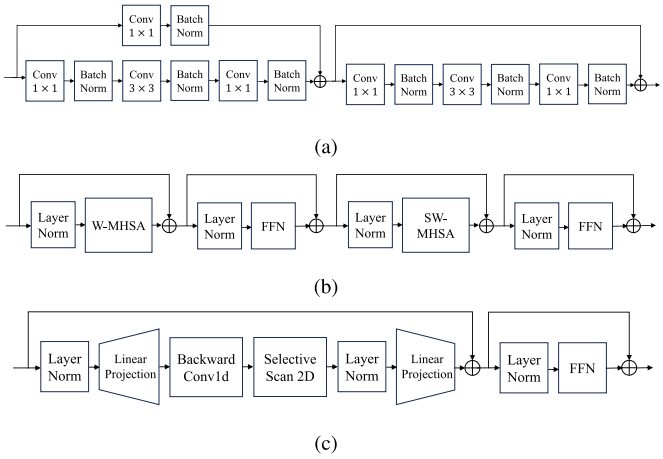


Fig. 7. Illustration of different representation learners used in the modularized Siamese encoder. The ResNet block consists of several convolutional layers with skip connections. The Swin Transformer block uses a window multihead self-attention (W-MHSA) and a subsequently shifted window multihead self-attention (SW-MHSA) to capture the global dependencies and an MLP layer to enable channelwise interactions. The VMamba block applies selective scan 2-D to capture the contextual information and also an MLP layer for channel mixing. (a) ResNet blocks. (b) Swin Transformer blocks. (c) VMamba blocks.

features from bitemporal images. This paradigm lacks scalability and adaptability for different change detection scenarios. To address this limitation, we propose a modularized Siamese encoder to build a more flexible change detection model. This encoder uses a weight-shared universal vision backbone with modularized representation learners that can be adjusted to different scales and incorporates various representation learning strategies. Consequently, our modularized Siamese encoder can adapt to heterogeneous remote sensing datasets with diverse scales and spectral characteristics.

Over the past decade, numerous vision backbones have been developed to enhance the efficacy of learning deep representations. These backbones can be broadly categorized into CNNs, transformer-based models, and Mamba-based models, reflecting their architecture and chronological development. As shown in Table I, these backbones use convolutional networks, self-attention mechanisms, and selective state space models to learn deep representations, respectively. Furthermore, Fig. 7 illustrates the architectures of the representation learners in these backbones. Each type of backbone has a distinct focus: CNNs concentrate on localized image areas, self-attention models capture long-range dependencies among all feature elements, and selective state space models reduce computational complexity for context modeling with a selective scan 2-D mechanism.

In this article, we select one representative model for each type of backbone: ResNet [47] for CNNs, Swin Transformer [48] for transformer-based models, and VMamba [49] for Mamba-based models. These vision backbones consist of four stages of representation learners, and we construct a modularized weight-shared Siamese structure to integrate these models to extract bitemporal features simultaneously from remote sensing images, as shown in Fig. 8. N_1, N_2,

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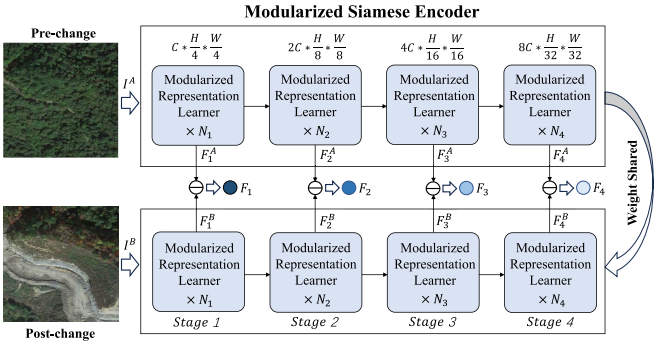


Fig. 8. Illustration of the modularized Siamese encoder. The shapes of the output feature map of different levels are indicated on the top of the representation of learners.

N_3, and N_4 denote the number of blocks of the modularized representation learners at each stage of the encoder, which can be adjusted to construct the backbones with different scales.

Let I^A and I^B be the pre- and postchange remote sensing images, respectively. The pre- and postchange deep features of four different levels, $l = \{1, 2, 3, 4\}$, are extracted by the modularized Siamese encoder. These features are denoted as $\{F_l^A\}_{l=1}^4$ and $\{F_l^B\}_{l=1}^4$, where F_l^A and F_l^B refer to the deep features at level l for the pre- and postchange images, respectively. After that, the multiscale bitemporal deep features are subtracted into feature differences $\{F_l\}_{l=1}^4$ as follows:

$$F_l = F_l^A - F_l^B. \tag{1}$$

Since these vision backbones have been widely adopted in a range of AI4EO applications, our objective is to conduct a comprehensive evaluation of these backbones for remote sensing change detection and validate the effectiveness of the proposed ChangeFFT module on various representation learners. Furthermore, we select three variants with different scales of parameters for each vision backbone to evaluate their scalability in the change detection task.

B. Change-Aware Fast Fourier Transform

Currently, most encoder–decoder-based change detection approaches directly use feature differences as change representations to reconstruct a change map in the decoder. However, the subtraction operation for fusing bitemporal deep features can disrupt the well-learned deep representations due to the complex spatiotemporal relationships of the land surfaces, leading to challenges in distinguishing changes in the decoder. To overcome this challenge, we propose a ChangeFFT module that effectively learns change-aware representations from multilevel feature differences in the frequency domain.

Learning in the frequency domain has proven to be an efficient technique for capturing channelwise dependencies of deep features [50]. By applying a fast Fourier transform (FFT), features are transformed from the time domain to the frequency domain, where change-aware representations can be precisely captured within the spectrum of feature differences. Let $\{F_l\}_{l=1}^4 \in \mathbb{R}^{C_l \times H_l \times W_l}$ be the differenced features input, where C_l, H_l, and W_l are the number of channels, height, and width of the lth level of feature differences, respectively. The FFT can be calculated as follows:

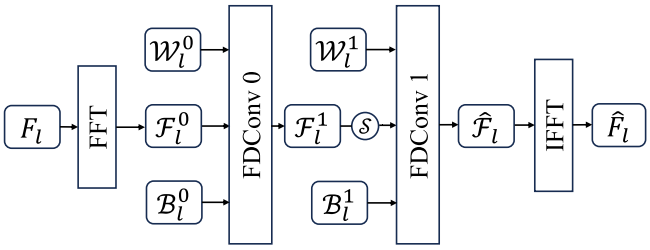


Fig. 9. Illustration of the ChangeFFT module. The $\mathcal{S}$ denotes SiLU activation function.

$$\begin{aligned}\mathcal{F}_l(k) &= \int_D F(x) e^{-j2\pi kx} dx \\ &= \int_D F_l(x) \cos(2\pi kx) dx + j \int_D F_l(x) \sin(2\pi kx) dx \\ &= \mathcal{F}_l^R + j\mathcal{F}_l^I \end{aligned} \tag{2}$$

where k is the frequency variable, and x indexes the spatial coordinates. $\mathcal{F}_l^R$ and $\mathcal{F}_l^I$ is the real and imaginary parts of the transformed feature $\mathcal{F}_l$ at the lth level, respectively. Subsequently, two convolutional layers are applied to enable a change-aware representation learning in the frequency domain. According to the convolutional theorem, the Fourier transform of a convolution of two tensors equals the pointwise product of their Fourier transforms, as follows:

$$\text{FFT}\{H * G\} = \mathcal{H} \cdot \mathcal{G} \tag{3}$$

where $*$ denotes the convolutional operator, and $\mathcal{H}$ and $\mathcal{G}$ are the Fourier transforms of tensors H and G, respectively. On this basis, a convolutional layer in the frequency domain (FDConv) can be initialized with learnable parameters $\mathcal{W}$ and $\mathcal{B}$ in complex representing the weights and biases, respectively. Let $\mathcal{F}$ be a feature spectrum, the FDConv can be computed as follows:

$$\begin{aligned}&\text{FDConv}(\mathcal{F}) \\ &= \text{FFT}(F * W + B) = \mathcal{F} \cdot \mathcal{W} + \mathcal{B} \\ &= \left(\mathcal{F}^R\mathcal{W}^R - \mathcal{F}^I\mathcal{W}^I + \mathcal{B}^R\right) + j\left(\mathcal{F}^R\mathcal{W}^I + j\mathcal{F}^I\mathcal{W}^R + \mathcal{B}^I\right)\end{aligned} \tag{4}$$

where the superscripts R and I indicate the real and imaginary elements of a complex tensor, respectively. As shown in Fig. 9, we apply two consecutive FDConvs to ensure a comprehensive change-aware representation learning for each level of the feature differences.

The FDConv can be viewed as a channel attention mechanism that learns the attention weights from the spectrum of feature differences in the frequency domain, which can be more effective in representing the channelwise features than the global average pooling used in the traditional squeeze-and-excitation channel attention block. In the meantime, the number of parameters of the ChangeFFT module for each level is restricted to the amount of $C_l \cdot C_l + C_l$, which is relatively lightweight compared with other complicated channel-wise feature enhancement methods such as multilevel

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TABLE II
SUMMARY OF CHANGE DETECTION DATASETS IMPLEMENTED IN THE UCD FRAMEWORK

Dataset	#Patches	Scenario	Location	Sensor	Resolution
CLCD [43]	2400	Cropland	Guangdong, China	Gaofen-2	0.5m-2m
EGY-BCD [42]	6091	Building	Egypt	Google Earth	0.25m
GVLM-CD [44]	7496	Landslide	Global	Google Earth	0.59m
LEVIR-CD [41]	10192	Building	Texas, USA	Google Earth	0.5m
SYSU-CD [28]	20000	Urban	Hong Kong, China	Aerial Image	0.5m
MineNetCD (Ours)	71711	Mining	Global	Google Earth	1.2m

perception (MLP). Our calculation indicates that the incorporation of the ChangeFFT module will only increase by about 10% of the model parameters, indicating its ability to learn the change-aware representations in a balanced performance.

After change-aware representation learning in two stages of FDConv, the obtained change representations are transformed back to the time domain with an inversed FFT (IFFT) function, as follows:

$$\hat{F}_l(x) = \int_D \hat{\mathcal{F}}_l(k) e^{j2\pi kx} dk = \int_D \left(\hat{\mathcal{F}}_l^R(k) + j\hat{\mathcal{F}}_l^I(k)\right) e^{j2\pi kx} dk \quad (5)$$

where $\hat{\mathcal{F}}_l(k)$ is the spectrum output of the second stage of FDConv, as illustrated in Fig. 9.

Note that the FFT and IFFT are performed along the channel dimension of the multiscale features. In conclusion, the multilevel change-aware representations $\{\hat{F}_l\}_{l=1}^4$ are obtained from feature differences $\{F_l\}_{l=1}^4$ in the ChangeFFT module.

C. Change Decoder

We use a multiscale semantic decoder, known as UperNet [51], to progressively fuse and upsample the multiscale change-aware representations into a full-resolution change map. The UperNet has been widely used as a plug-and-play semantic segmentor due to its superior compatibility with various vision backbones, which we considered also suitable for integrating with the modularized backbone and ChangeFFT module in this study.

The UperNet change decoder processes multiscale change-aware representations through a hierarchical approach that combines a PPM and an FPN. Initially, the PPM extracts global context information by applying pooling operations at multiple scales to the deepest change representations, creating a set of pooled features that capture various levels of contextual information. These pooled features are then concatenated and upsampled to match the original resolution. Subsequently, the FPN aggregates features from different layers of the backbone network, progressively refining and merging them to produce high-resolution feature maps. These multiscale features from the PPM and FPN are then fused to generate a comprehensive feature representation, which is finally passed through convolutional layers to produce the change map, effectively leveraging both local details and global context for accurate change detection.

D. Loss Function

We use a pixelwise binary cross-entropy (BCE) loss $\mathcal{L}_{\text{BCE}}$ to train the MineNetCD model. The BCE loss measures the discrepancy between the predicted change probabilities and the ground truths for each pixel as follows:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right] \quad (6)$$

where i indexes a pixel, and N is the number of pixels. y_i denotes a binary ground truth (0 for unchanged, 1 for changed) of pixel i, and p_i is the predicted probability that pixel i belongs to the changed class.

IV. UCD FRAMEWORK

To promote deep-learning-based change detection tasks for broader utilization by RS experts, we develop a UCD framework based on the Huggingface platform with an open science prospect. As shown in Tables III and II, the UCD framework currently includes 20 change detection methods and six change detection datasets, including the MineNetCD dataset and the model proposed in this article. The UCD framework allows the user to simply specify the change detection method and dataset in the configuration file, on which the training and testing experiments can be automatically conducted without any professional skills. As a result, the change detection model can be initialized or loaded using only one code line, respectively, as follows:

```
model=model.from_config(CONFIG)
model=model.from_pretrained(MODEL_ID)
```

where CONFIG and MODEL_ID represent the model configurations and model paths, respectively. In addition, we leverage the Huggingface platform to facilitate dataset access and model loading within a cloud-based framework. Specifically, we have uploaded the preprocessed datasets and pretrained model weights to the Huggingface Hub, allowing them to be automatically loaded into the UCD framework by specifying their dataset and model IDs in the configuration file. As a pioneering effort, we have made available more than 70 model weights, pretrained from 20 different approaches on six change detection datasets, on our Huggingface Hub https://huggingface.co/ericyu. These pretrained models can be used for future research to reproduce the results on these datasets, eliminating the need for retraining and enhancing the transparency of the experiments.

The UCD is also designed with high compatibility that allows the users to easily integrate their own resources into the

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TABLE III
SUMMARY OF CHANGE DETECTION APPROACHES IMPLEMENTED IN THE UCD FRAMEWORK

Methods	Backbone	#Params	FLOPS	Source
A2Net [52]	VGG16	3.60M	2.86G	IEEE TGRS 2023
AFCF3D [53]	ResNet18	16.83M	29.54G	IEEE TGRS 2023
BIT [30]	ResNet18	11.39M	8.28G	IEEE TGRS 2021
CGNet [54]	VGG16	37.18M	81.66G	IEEE JSTARS 2023
ChangeFormer [32]	MIT	39.13M	129.7G	IEEE IGARSS 2022
DMINet [27]	ResNet18	6.44M	16.16G	IEEE TGRS 2021
DTCDSCN [55]	-	39.17M	10.72G	IEEE GRSL 2020
FC-EF [56]	-	1.29M	2.92G	IEEE ICIP 2018
FCNPP [57]	-	14.56M	43.10G	IEEE GRSL 2019
HCGMNet [58]	VGG16	45.13M	301.82G	IEEE IGARSS 2023
ICIFNet [59]	ResNet18	24.64M	22.97G	IEEE TGRS 2022
MSPSNet [60]	-	2.11M	13.89G	IEEE TGRS 2021
RDPNet [61]	-	1.62M	1.63G	IEEE TGRS 2022
ResUnet [62]	-	12.59M	28.51G	IEEE IGARSS 2021
SiamUnet-Conc [56]	-	1.47M	4.55G	IEEE ICIP 2018
SiamUnet-Diff [56]	-	1.29M	3.99G	IEEE ICIP 2018
SNUNet [63]	-	11.48M	43.6G	IEEE GRSL 2021
TFI-GR [64]	ResNet18	27.06M	9.09G	IEEE TGRS 2022
TinyCD [65]	EfficientNet-b4	0.27M	1.44G	NCA 2023
MineNetCD (Ours)	SwinT-Tiny	57.81M	63.28G	This Paper

framework. We provide guidance and some tools for the users to preprocess and adapt their own datasets and models into standard formations that can be used in the UCD framework. We hope this new approach to the UCD framework will motivate future change detection researchers to participate in these open science activities, as well as provide an opportunity for everyone with interest to try change detection without any professional skills.

V. EXPERIMENTS

A. Dataset Preparation

To facilitate the training and testing of the baseline models, the bitemporal remote sensing image pairs from MineNetCD are cropped into a size of 256 × 256. We randomly selected 60%, 30%, and 10% of mining sites as the training, testing, and validation sets, respectively. Finally, the training, testing, and validation sets include a total of 47 743, 19 355, and 4613 patches, respectively.

In accordance with the open-science paradigm of the proposed UCD framework, the MineNetCD dataset can be easily accessed and integrated into the framework via https://huggingface.co/datasets/ericyu/MineNetCD256.

B. Comparison Approaches

The proposed MineNetCD baseline model is compared against 19 deep-learning-based change detection models implemented in the UCD framework: a lightweight network with progressive aggregation and supervised attention (A2Net) [52], an adjacent-level feature cross-fusion with 3-D CNN (AFCF3D) [53], a bitemporal image transformer (BIT) network [30], a change guiding network (CGNet) [54], a transformer-based Siamese network for change detection (ChangeFormer) [32], a dual-branch multilevel intertemporal network (DMINet) [27], a dual-task constrained deep Siamese convolutional network (DTCDSCN) [55], fully convolutional Siamese networks for change detection (FC-EF) [56], a fully convolutional network within pyramid pooling (FCNPP) [57], a hierarchical change guiding map network (HCGMNet) [58], an intrascale cross-interaction and interscale feature fusion network (ICIFNet) [59], a deep multiscale Siamese network with parallel convolutional structure and self-attention (MSPSNet) [60], a region detail preserving network (RDPNet) [61], a residual U-Net (ResUnet) [62], a fully convolutional Siamese concatenated U-Net (SiamUnet-Conc) [56], a fully convolutional Siamese difference U-Net (SiamUnet-Diff) [56], an integrated Siamese network and nested U-Net (SNUNet) [63], a network via temporal feature interaction and guided refinement (TFI-GR) [64], and a lightweight and effective change detection model called TinyCD [65].

C. Experimental Settings

The Adam optimizer was used for model training with an initial learning rate of $1e-4$, and the batch size was set to 32 for each GPU. Moreover, a cosine annealing scheduler is adopted to improve the convergence of the model by gradually reducing the learning rate to $1e-7$.

In our experiments, we used the default hyperparameter values for the comparison methods, as specified in their original publications. We set the random seed to 8888 to ensure the reproducibility of the experiments. All the experiments were executed with an HPC cluster featuring four NVIDIA Tesla A100 GPUs and a 128-core CPU. Moreover, we use the Accelerate [66] package to optimize the distributed learning system for faster computation.

D. Evaluation Metrics

We used five commonly used evaluation metrics for quantitative comparison: overall accuracy (OA), precision (Pre), recall (Rec), F1-score, and change-class intersection over union (cIoU). The OA indicates the proportion of correctly detected pixels out of all samples. Precision measures the ratio of correctly detected changed pixels to the total number

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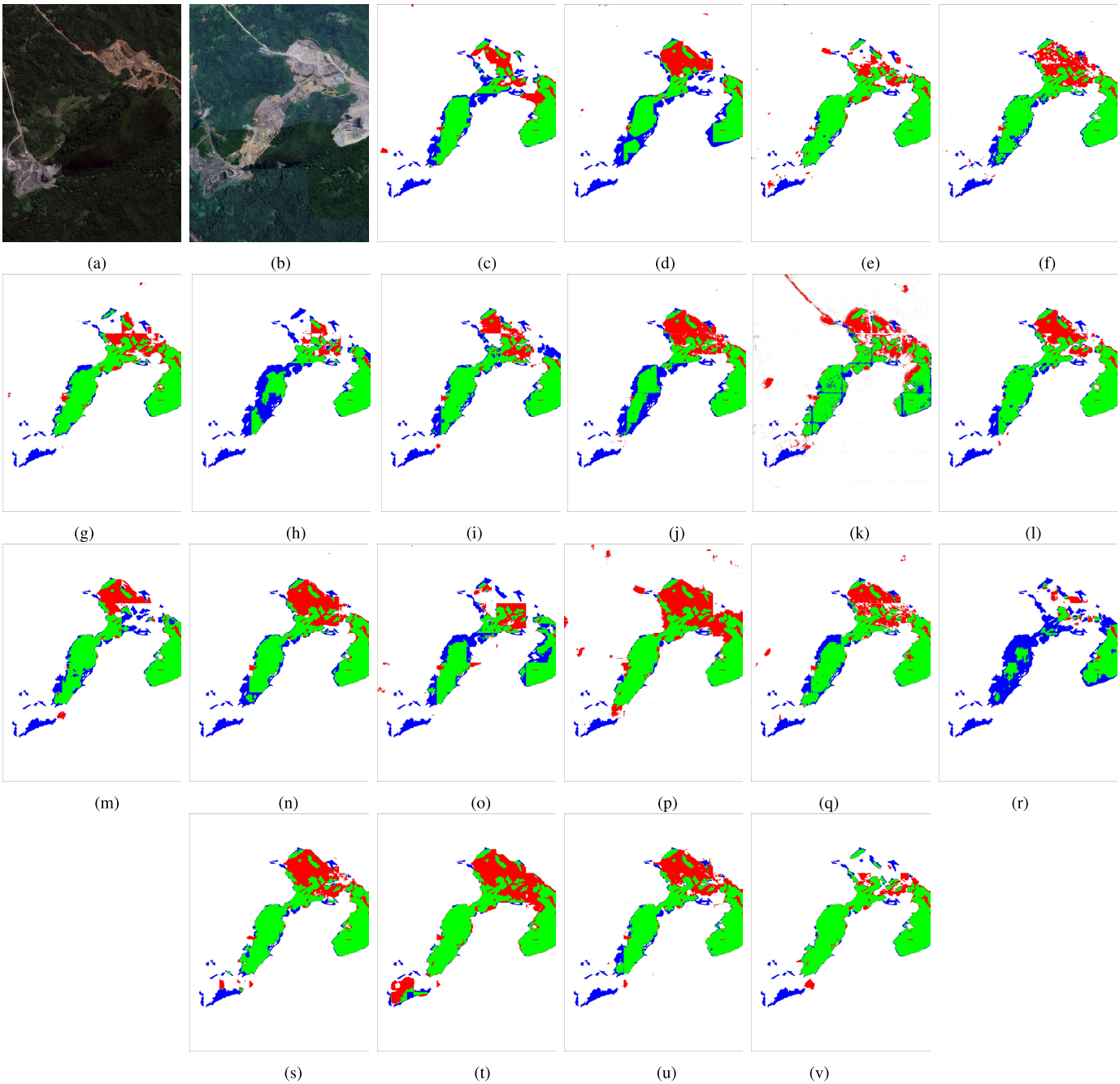


Fig. 10. Qualitative comparison for the site "Brunei_01" of the MineNetCD dataset. The rendered colors represent true positives (green), true negatives (white), false positives (red), and false negatives (blue). (a) Prechange. (b) Postchange. (c) A2Net. (d) AFCF3D. (e) BIT. (f) CGNet. (g) ChangeFormer. (h) DMINet. (i) DTCDSCN. (j) FC-EF. (k) FCNPP. (l) HCGMNet. (m) ICIFNet. (n) MSPSNet. (o) RDPNet. (p) ResUnet. (q) SiamUnet-Conc. (r) SiamUnet-Diff. (s) SNUNet. (t) TFI-GR. (u) TinyCD. (v) MineNetCD (Ours).

of pixels identified as changed in the detection map. The recall represents the ratio of correctly detected changed pixels among all pixels that should be identified as changed. The F1-score, calculated as the harmonic mean of precision and recall, accounts for class imbalance in change detection. The cIoU quantifies the exact overlap between the detected change map and the ground truth. They are calculated according to the following equations:

$$\mathrm{OA} = (\mathrm{TP} + \mathrm{TN})/(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}) \tag{7}$$

$$\mathrm{Pre} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP}) \tag{8}$$

$$\mathrm{Rec} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN}) \tag{9}$$

$$\mathrm{F1} = 2\mathrm{TP}/(2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}) \tag{10}$$

$$\mathrm{cIoU} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN}) \tag{11}$$

where TP is the number of true positive pixels that are correctly detected as changed, TN is the number of true negative pixels that are correctly detected as detected unchanged, FP is the number of false-positive pixels that are false alarms, and FN is the number of false-negative pixels, or missed changes.

E. Benchmark Results

1) Quantitative Results: Table IV displays the quantitative comparisons on the MineNetCD dataset. The results show

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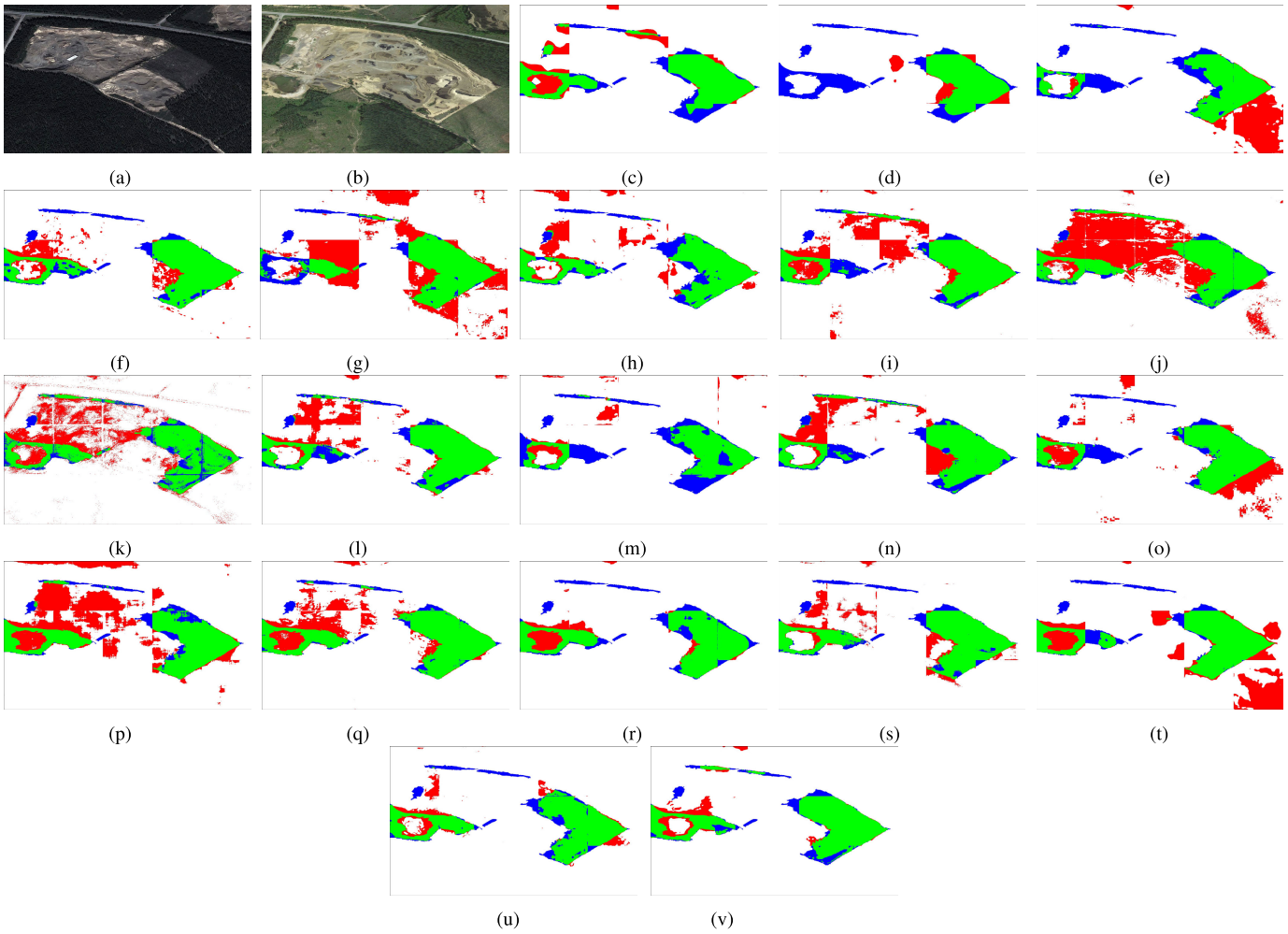


Fig. 11. Qualitative comparison for the site "Finland_01" of the MineNetCD dataset. The rendered colors represent true positives (green), true negatives (white), false positives (red), and false negatives (blue). (a) Prechange. (b) Postchange. (c) A2Net. (d) AFCF3D. (e) BIT. (f) CGNet. (g) ChangeFormer. (h) DMINet. (i) DTCDSCN. (j) FC-EF. (k) FCNPP. (l) HCGMNet. (m) ICIFNet. (n) MSPSNet. (o) RDPNet. (p) ResUnet. (q) SiamUnet-Conc. (r) SiamUnet-Diff. (s) SNUNet. (t) TFI-GR. (u) TinyCD. (v) MineNetCD (Ours).

TABLE IV

QUANTITATIVE COMPARISONS IN TERMS OF OA, PRE, REC, F1, AND CIOU ON THE MINENETCD DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Methods	OA	Precision	Recall	F1	cIoU
A2Net [52]	0.9188	0.7218	0.5766	0.6410	0.4718
AFCF3D [53]	0.8937	0.5768	0.5796	0.5782	0.4066
BIT [30]	0.9118	0.6732	0.5804	0.6233	0.4528
CGNet [54]	0.9004	0.6412	0.4770	0.5470	0.3765
ChangeFormer [32]	0.8705	0.4857	0.5160	0.5003	0.3337
DMINet [27]	0.8967	0.6266	0.4412	0.5177	0.3493
DTCDSCN [55]	0.8989	0.6197	0.5066	0.5573	0.3864
FC-EF [56]	0.8836	0.5628	0.3293	0.4154	0.2622
FCNPP [57]	0.8559	0.4022	0.3024	0.3452	0.2087
HCGMNet [58]	0.9076	0.6718	0.5222	0.5876	0.4161
ICIFNet [59]	0.8920	0.5966	0.4342	0.5024	0.3356
MSPSNet [60]	0.8998	0.6277	0.5041	0.5591	0.3880
RDPNet [61]	0.8774	0.5128	0.4818	0.4968	0.3305
ResUnet [62]	0.8671	0.4753	0.5512	0.5104	0.3427
SiamUnet-Conc [56]	0.8984	0.6469	0.4220	0.5108	0.3430
SiamUnet-Diff [56]	0.8960	0.7679	0.2478	0.3746	0.2305
SNUNet [63]	0.8927	0.5829	0.5124	0.5452	0.3750
TFI-GR [64]	0.8833	0.5257	0.7310	0.6116	0.4406
TinyCD [65]	0.8999	0.6250	0.5152	0.5648	0.3936
MineNetCD (Ours)	0.9251	0.7120	0.6814	0.6963	0.5343

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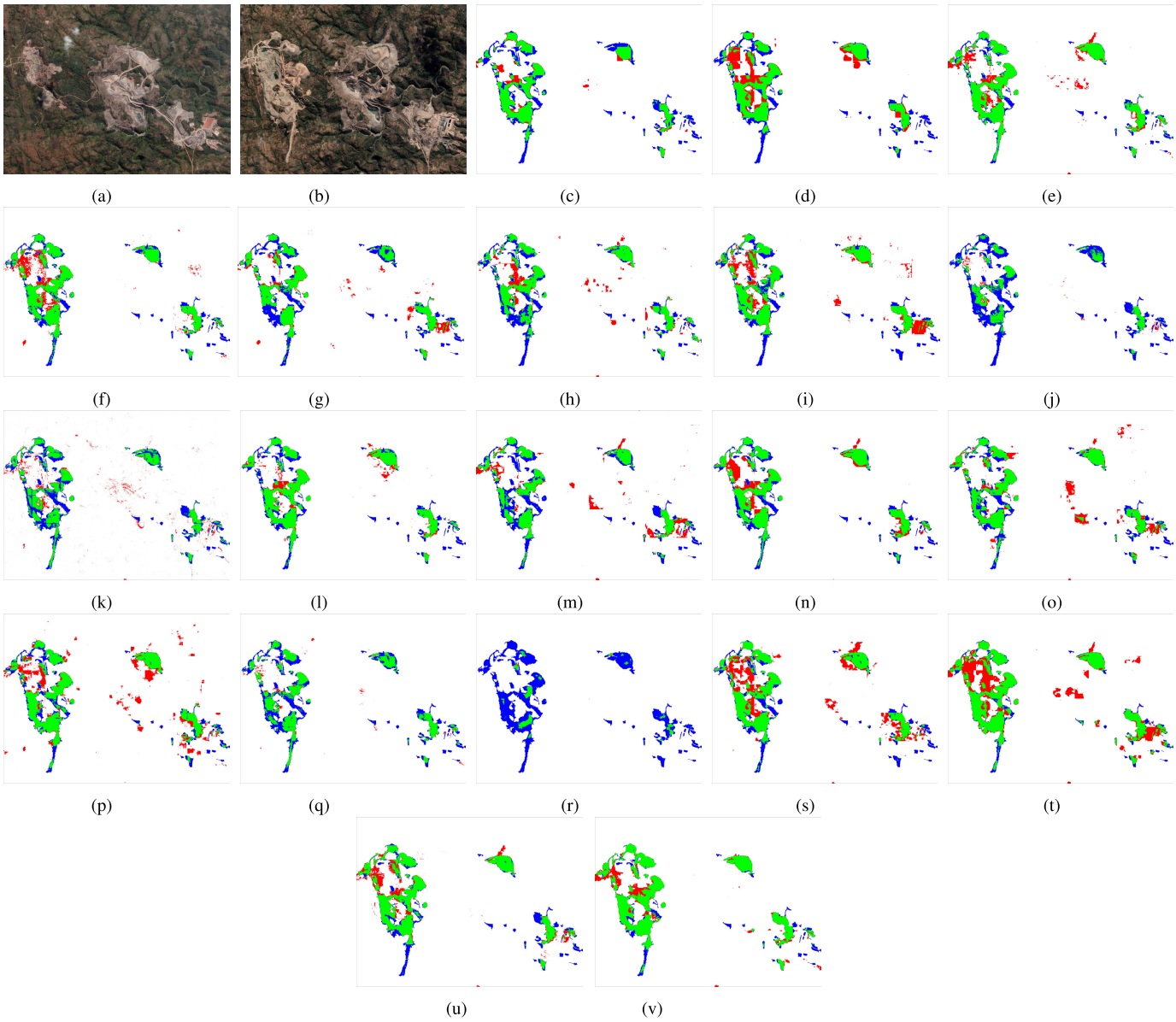


Fig. 12. Qualitative comparison for the site "Congo_06" of the MineNetCD dataset. The rendered colors represent true positives (green), true negatives (white), false positives (red), and false negatives (blue). (a) Prechange. (b) Postchange. (c) A2Net. (d) AFCF3D. (e) BIT. (f) CGNet. (g) ChangeFormer. (h) DMINet. (i) DTCDSCN. (j) FC-EF. (k) FCNPP. (l) HCGMNet. (m) ICIFNet. (n) MSPSNet. (o) RDPNet. (p) ResUnet. (q) SiamUnet-Conc. (r) SiamUnet-Diff. (s) SNUNet. (t) TFI-GR. (u) TinyCD. (v) MineNetCD (Ours).

that the MineNetCD dataset is challenging for current change detection methods, with an average F1 score of less than 0.65 and a cIoU of less than 0.48. Although some methods achieve promising recall values, they suffer from low precision, which leads to a lot of false alarms. On the contrary, the proposed model can achieve an F1 score of 0.6963 and a cIoU of 0.5343, which gains an improvement of at least 0.0553 and 0.0625 in terms of F1 score and cIoU, respectively, compared to all the competitors. The overall results demonstrate the superiority and effectiveness of the proposed MineNetCD model.

2) Qualitative Results: Figs. 10–12 show examples of MineNetCD results obtained using various comparative methods. In addition, we present the results for original small patches sized 256 × 256 to enable a comprehensive quantitative comparison. The visualizations reveal that most comparison methods struggle to accurately detect the correct changed areas while avoiding false alarms.

Based on these representative results, the proposed MineNetCD method demonstrates advantages in several aspects.

1) *Better Discrimination of Changes:* In the first five rows of Fig. 13, most comparison methods fail to distinguish the changes in bitemporal images, resulting in change maps with numerous omitted changes (false negatives, shown in blue). Conversely, the MineNetCD method consistently produces accurate change maps from these challenging samples, highlighting its superior ability to differentiate changes in bitemporal images with heterogeneous spectral characteristics (e.g., rows one–three in Fig. 13) and subtle spectral differences (e.g., rows four and five in Fig. 13). This superior discrimination of

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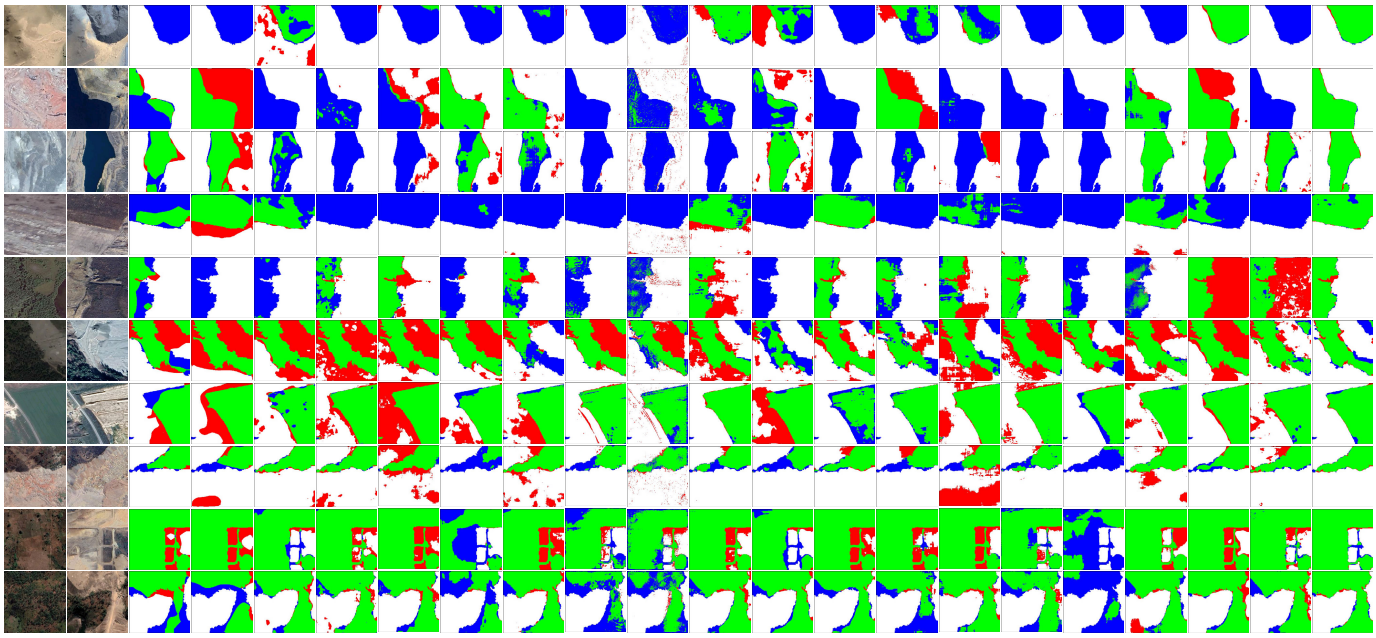


Fig. 13. Visual comparisons of the proposed method and the state-of-the-art approaches on the MineNetCD dataset. (a) Prechange images. (b) Postchange images. (c) A2Net. (d) AFCF3D. (e) BIT. (f) CGNet. (g) ChangeFormer. (h) DMINet. (i) DTCDSCN. (j) FC-EF. (k) FCNPP. (l) HCGMNet. (m) ICIFNet. (n) MSPSNet. (o) RDPNet. (p) ResUnet. (q) SiamUnet-Conc. (r) SiamUnet-Diff. (s) SNUNet. (t) TFI-GR. (u) TinyCD. (v) MineNetCD (Ours). The rendered colors represent true positives (green), true negatives (white), false positives (red), and false negatives (blue).

the MineNetCD model is attributed to the ChangeFFT module, which aligns spatiotemporal feature differences by unifying their spectra in the frequency domain to obtain robust change-aware representations.

2) *Fewer Pseudochanges:* Compared with other methods, the MineNetCD model significantly reduces pseudochanges—false alarms or irrelevant changes to mining development. In Fig. 11, change maps generated by other methods exhibit extensive pseudochanges (denoted as red patches). In the sixth row of Fig. 13, most methods predict pseudochanges in mining areas already present in the prechange image. In contrast, the change maps from the MineNetCD model exhibit fewer pseudochanges, benefiting from the ChangeFFT module's ability to filter out noise in bitemporal feature differences by processing their feature spectra in the frequency domain.

3) *More Precise Boundaries:* Mining change detection often involves irregularly shaped instances due to the nature of open-pit mining developments. It is crucial to accurately capture the extent of the mining area. In Figs. 10 and 12, and rows eight–ten of Fig. 13, the change maps predicted by the comparison methods appear to have incorrect and unclear instance edges, while the MineNetCD model managed to capture precise edges of change instances in mining areas. This precision is achieved through change-aware representation learning in the ChangeFFT module, which accurately extracts spatiotemporal dependencies in the frequency domain.

The changes occurring in the benchmark set are very diverse and relevant for both the monitoring of the mining activities and their relevance for the environment and the local communities. The changes occur in the pits themselves, where extraction occurs, but also in the waste piles and tailings, where wastes are deposited, or the different processing units surrounding the mines. This is important because the impacts of mining vary from commodity to commodity, processing types, and environmental conditions. Only a holistic approach to mining changes can provide sufficient information to monitor and address the different impacts of mining sites.

F. Comparisons of Different Backbone Models for Change Detection

To explore the influence of different representation learners on the performance of change detection, we apply three distinct branches of backbone models, each with three variants at different scales, to build the modularized Siamese encoder of the MineNetCD model. As shown in Fig. 14, the Swin Transformer achieves the best performance on MineNetCD in terms of F1 score and cIoU among all the scales of the backbone variants. In particular, the highest performance is achieved by the Swin Transformer Tiny, indicating that the deep features extracted by a larger self-attention-based backbone may be too abstract to distinguish changes in MineNetCD.

Furthermore, we leverage the GradCAM [67] technique to visualize the change-aware representations of different backbone variants, as shown in Fig. 15. It can be observed that the ResNet variants cannot extract effective features among the changed areas, as shown in the label. Furthermore, the heatmaps of the change-aware representations generated by the VMamba series have oversmooth boundaries, which cannot precisely characterize the extent of changed areas.

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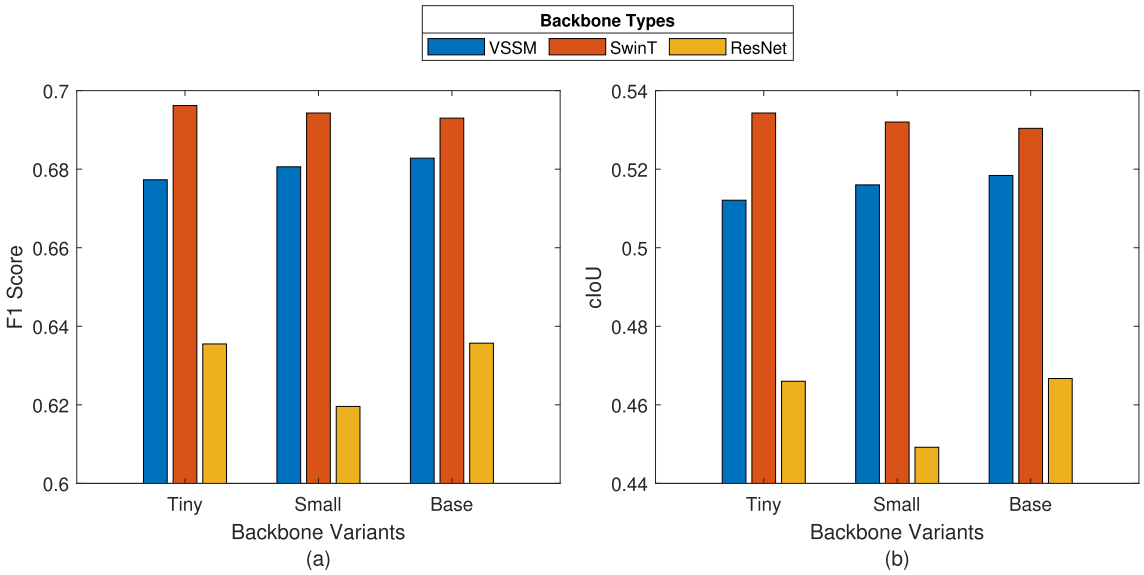


Fig. 14. Comparison results of MineNetCD with different backbone models, including VMamba, Swin Transformer (SwinT), and ResNet on the MineNetCD dataset. The tiny, small, and base scales of the ResNet model indicate ResNet-18, ResNet-50, and ResNet-101, respectively. (a) and (b) Results in terms of F1 score and cIoU metrics, respectively.

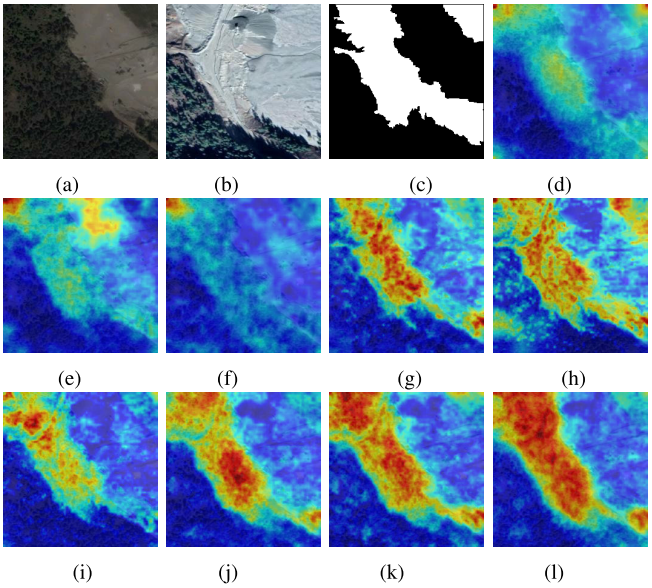


Fig. 15. GradCAM visualizations of change-aware representations using different backbone variants on MineNetCD. (a) Prechange. (b) Postchange. (c) Label. (d) ResNet-18. (e) ResNet-50. (f) ResNet-101. (g) Swin-T. (h) Swin-S. (i) Swin-B. (j) VMamba-T. (k) VMamba-S. (l) VMamba-B.

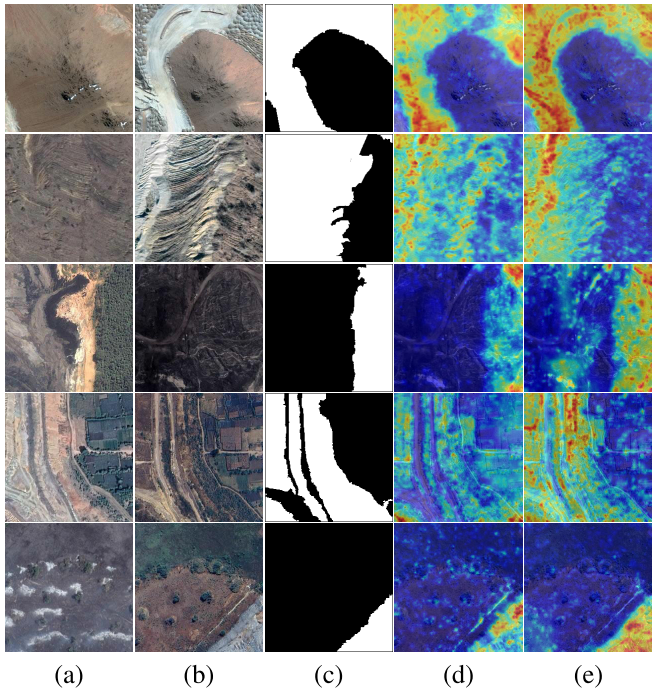


Fig. 16. Comparison of GradCAM visualizations of feature differences and change-aware representations. (a) Prechange images. (b) Postchange images. (c) Labels. (d) Bitemporal deep feature differences. (e) Change-aware representations learned by ChangeFFT.

In contrast, the Swin Transformer series can precisely capture the change-aware representations with sharp boundaries, indicating its ability to process the bitemporal RS images into effective features with the ChangeFFT module.

G. Ablation Studies

To evaluate the effectiveness of the ChangeFFT module, we conducted extensive ablation studies to build the MineNetCD model without using the ChangeFFT module. In this case, the multiscale deep feature differences are directly fed into the UperNet-based change decoder to reconstruct a change map. Our experiments demonstrated that the ChangeFFT module improves performance in all settings based on various backbones, as shown in Table V. In particular, incorporating the ChangeFFT module improves up to 3.40% F1 score and 3.89% cIoU based on the Swin Transformer base backbone. The ChangeFFT module is also effective on the VMamba series and ResNet series, indicating that the ChangeFFT module possesses superior compatibility and the potential to become a plug-and-play module in more change detection approaches.

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TABLE V
PERFORMANCE IMPROVEMENTS OF CHANGEFFT BASED ON DIFFERENT BACKBONE VARIANTS

Backbones	Variants	F1	cIoU
VMamba	VMamba-S	+0.0163	+0.0184
	VMamba-T	+0.0055	+0.0061
	VMamba-B	+0.0234	+0.0262
Swin Transformer	Swin-T	+0.0340	+0.0389
	Swin-S	+0.0219	+0.0253
	Swin-B	+0.0226	+0.0258
ResNet	ResNet-18	+0.0098	+0.0106
	ResNet-50	+0.0098	+0.0100
	ResNet-101	+0.0085	+0.0089

We also display GradCAM visualization to compare the conventional feature difference methods and the change-aware representations based on the Swin Transformer base backbone, as shown in Fig. 16. It can be seen in the results that the ChangeFFT module significantly refines the feature difference and obtains change-aware representations highly focused on the changed areas. Specifically, the change-aware representations have a more accurate mapping of ground instances and reduce the noises within the feature differences.

VI. CONCLUSION

In this article, we introduce MineNetCD, a comprehensive benchmark for global mining change detection aimed at promoting the sustainability goals of the extractive industry. MineNetCD involves a global mining change detection dataset, a ChangeFFT module-based baseline model, and a UCD framework. The dataset includes bitemporal high-resolution remote sensing images and pixel-level annotations of changed areas from 100 mining sites worldwide. Our baseline model, using the ChangeFFT module, achieves better than state-of-the-art performance by enabling change-aware representation learning in the frequency domain. The UCD framework features over 70 pretrained models based on six change detection datasets, democratizing access to deep-learning-based change detection methods without requiring specialized expertise.

This work provides a robust solution for mining change detection, which is crucial for environmental impact assessments in the mining industry using geospatial data. We anticipate that MineNetCD will drive advancements in sustainable metal sourcing and inspire future research focused on environmentally responsible mining practices. Looking ahead, we encourage researchers to develop more sophisticated mining monitoring schemes based on the MineNetCD benchmark. First, expanding the number of mining sites studied will allow for a more comprehensive assessment of mining activities across various countries. Second, we advocate for the development of more advanced deep learning algorithms with enhanced generalization capabilities to improve accuracy in mining change detection across highly heterogeneous global mining sites. Finally, we invite more researchers to join our open science initiative, using and contributing to the UCD framework in their future studies.

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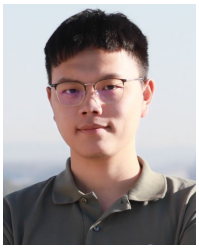
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Weikang Yu (Graduate Student Member, IEEE) received the B.Eng. degree from Beihang University, Beijing, China, in 2020, and the M.Phil. degree from The Chinese University of Hong Kong, Shenzhen, in 2022. He is currently pursuing the Ph.D. degree with the Machine Learning Group (FWGE-L), Helmholtz Institute Freiberg for Resource Technology, Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Freiberg, Germany.

He also works as a Ph.D. Student with the Chair of Data Science in Earth Observation with Technical University of Munich (TUM), Munich, Germany. His research interests include remote sensing, image processing, and machine learning.



Xiaokang Zhang (Senior Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2018.

From 2019 to 2022, he was a Postdoctoral Research Associate with The Hong Kong Polytechnic University, Hong Kong, and The Chinese University of Hong Kong, Shenzhen, Shenzhen, China. He is currently a specially appointed Professor with the School of Information Science and Engineering, Wuhan University of Science and Technology, Wuhan. He has authored or co-authored more than 40 scientific publications in international journals and conferences. His research interests include remote sensing image analysis, pattern recognition, and machine learning.



Richard Gloaguen received the Ph.D. degree "Doctor Communitatis Europae" in marine geosciences from the University of Western Brittany, Brest, France, in collaboration with the Royal Holloway University of London, London, U.K., and Göttingen University, Göttingen, Germany, in 2000.

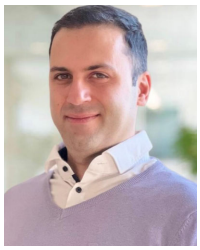
He is a leading researcher in the fields of computer vision, machine learning, and spectroscopy, particularly in the context of imaging sensors. His work at the Helmholtz Institute Freiberg, Freiberg, Germany, for Resource Technology focuses on integrating advanced technologies to enhance the characterization of complex materials. His contributions are notable in advancing methodologies that combine computer vision and machine learning with traditional spectroscopic techniques, thereby pushing the boundaries of imaging technologies.



Xiao Xiang Zhu (Fellow, IEEE) received the master's (M.Sc.), doctor of engineering (Dr.-Ing.), and her "Habilitation" degrees in the field of signal processing from Technical University of Munich (TUM), Munich, Germany, in 2008, 2011, and 2013, respectively.

She is the Chair Professor for Data Science in Earth Observation with TUM. She was the Founding Head of the Department "EO Data Science" at the Remote Sensing Technology Institute, German Aerospace Center (DLR), Wessling, Germany. Since May 2020, she has been the PI and Director of the International Future AI Laboratory "AI4EO—Artificial Intelligence for Earth Observation: Reasoning, Uncertainties, Ethics and Beyond," Munich. Since October 2020, she also serves as the Director of the Munich Data Science Institute (MDSI), TUM. From 2019 to 2022, she has been a Co-Coordinator of the Munich Data Science Research School (www.mu-ds.de) and the Head of the Helmholtz Artificial Intelligence—Research Field "Aeronautics, Space and Transport." She was a Guest Scientist or a Visiting Professor with the Italian National Research Council (CNR-IREA), Naples, Italy; Fudan University, Shanghai, China; the University of Tokyo, Tokyo, Japan; and University of California at Los Angeles, Los Angeles, CA, USA, in 2009, 2014, 2015, and 2016, respectively. She is currently a Visiting AI Professor at ESA's Phi-Laboratory, Frascati, Italy. Her main research interests include remote sensing and Earth observation, signal processing, machine learning, and data science, with their applications in tackling societal grand challenges, e.g., global urbanization, UN's SDGs, and climate change.

Dr. Zhu is a fellow of the Academia Europaea (the Academy of Europe), AAIA, and ELLIS. She has been a member of Young Academy (Junge Akademie/Junges Kolleg) at the Berlin-Brandenburg Academy of Sciences and Humanities and the German National Academy of Sciences Leopoldina and the Bavarian Academy of Sciences and Humanities. She serves on the Scientific Advisory Board in several research organizations, among others the German Research Center for Geosciences (GFZ) from 2020 to 2023, and the Potsdam Institute for Climate Impact Research (PIK). She is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and *Pattern Recognition* and served as the Area Editor responsible for special issues of *IEEE Signal Processing Magazine* from 2021 to 2023.



Pedram Ghamisi (Senior Member, IEEE) received the Ph.D. degree in electrical and computer engineering from the University of Iceland, Reykjavik, Iceland, in 2015.

He currently serves as the Head of the Machine Learning Group, Helmholtz Institute (HZDR-HIF), Dresden, Germany, and a Visiting Professor with Lancaster University, Lancaster, U.K. He has previously held positions as a Senior PI, a Research Professor, and a Group Leader of AI4RS at the Institute of Advanced Research in Artificial Intelligence (IARAI), Vienna, Austria. His research interests primarily revolve around deep learning, particularly in the domain of remote sensing applications.

Dr. Ghamisi has achieved more than ten international awards and recognitions, including the IEEE-GRSS Highest Impact Paper Awards for 2020 and 2024, the Winner of the Data Fusion Contest 2017 organized by IEEE GRSS, the IEEE Mikio Takagi Prize for 2013, and the 2017 Best Reviewer Prize of IEEE GRSS. In addition, he was named one of the top 1% of most cited researchers, as published by Clarivate since 2021. He has also been the recipient of multiple scholarships and grants, such as the Alexander von Humboldt Fellowship in 2015, the High Potential Program (HPP) Team Leadership in 2018, the AI for Remote Sensing grant in 2020, and the 3D-ABC of the Helmholtz Foundation Models Initiative in 2024. For more information, visit http://www.ai4rs.com.

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